# K-Means as a Matrix Factorization Problem

## 1 Introduction

K-means clustering is traditionally formulated as minimizing within-cluster variance. However, it can also be expressed as a constrained matrix factorization problem. This formulation reveals deep connections between clustering, low-rank approximation, spectral methods, and principal component analysis.

## 2 Classical K-Means Objective

Let:

- $X \in \mathbb{R}^{n \times d}$ be the data matrix.
- Each row $x_i \in \mathbb{R}^d$ represents a data point.
- $k$ be the number of clusters.
- $C_1, \ldots, C_k$ be the clusters.
- $\mu_j \in \mathbb{R}^d$ be the centroid of cluster $j$.

The classical k-means objective is:

$$\min_{\{C_j\}} \sum_{j=1}^{k} \sum_{x_i \in C_j} \|x_i - \mu_j\|^2. \tag{1}$$

## 3 Matrix Representation of Cluster Assignments

Define the cluster assignment matrix:

$$H \in \{0,1\}^{n \times k}$$

where:

$$H_{ij} = \begin{cases} 1 & \text{if data point } i \text{ belongs to cluster } j \\ 0 & \text{otherwise.} \end{cases}$$

Each data point belongs to exactly one cluster, so:

$$H\mathbf{1}_k = \mathbf{1}_n.$$

Define the centroid matrix:

$$M \in \mathbb{R}^{k \times d}$$

whose $j$-th row is $\mu_j^T$.

## 4 Why $X \approx HM$? (Detailed Explanation)

The key insight is that in k-means, each data point is approximated by the centroid of the cluster to which it belongs.

Let us examine the matrix product:

$$HM.$$

Consider row $i$ of $HM$:

$$(HM)_i = \sum_{j=1}^{k} H_{ij} M_j.$$

Because each row of $H$ contains exactly one 1 and the rest zeros, there exists exactly one index $c(i)$ such that:

$$H_{i,c(i)} = 1.$$

Therefore,

$$(HM)_i = M_{c(i)}.$$

But $M_{c(i)}$ is precisely the centroid of the cluster assigned to data point $i$.
Thus:

$$(HM)_i = \mu_{c(i)}.$$

Hence, the matrix product $HM$ constructs a new matrix where each row equals the centroid of the cluster assigned to that data point.

Therefore:

$$X \approx HM$$

means:

Each data point $x_i$ is approximated by the centroid of its assigned cluster.

### Geometric Interpretation

The matrix $HM$ represents a reconstruction of the data where:

- All points within a cluster collapse to their centroid.
- The dataset becomes piecewise constant across clusters.

Thus, k-means is finding the best approximation of $X$ using only $k$ distinct representative vectors (the centroids).

## 5 Matrix Factorization Formulation

The k-means objective can now be written as:

$$\min_{H,M} \|X - HM\|_F^2 \tag{2}$$

subject to:

- $H \in \{0,1\}^{n\times k}$
- $H\mathbf{1}_k = \mathbf{1}_n$

This is a low-rank matrix factorization problem with discrete constraints.

## 6 Solving for $M$ Given $H$

Fix $H$ and minimize over $M$:

$$\|X - HM\|_F^2.$$

Take derivative with respect to $M$:

$$\frac{\partial}{\partial M}\|X - HM\|_F^2 = -2H^T(X - HM).$$

Setting to zero:

$$H^T X = H^T HM.$$

Solving:

$$M = (H^T H)^{-1} H^T X.$$

Since $H^T H$ is diagonal with cluster sizes:

$$(H^T H)_{jj} = |C_j|,$$

we obtain:

$$\mu_j = \frac{1}{|C_j|}\sum_{x_i \in C_j} x_i.$$

Thus, the optimal $M$ recovers the usual cluster means.

## 7 Reduced Objective in $H$ Only

Substituting $M$ back:

$$\min_H \|X - H(H^T H)^{-1} H^T X\|_F^2.$$

Define:

$$P_H = H(H^T H)^{-1} H^T.$$

Then:

$$X \approx P_H X.$$

This shows that k-means performs a projection of $X$ onto a subspace defined by cluster indicator structure.

## 8 Trace Maximization Form

After algebraic simplification:

$$\max_H \operatorname{Tr}(H^T X X^T H (H^T H)^{-1}).$$

Define normalized indicator matrix:

$$F = H(H^T H)^{-1/2}.$$

Then:

$$F^T F = I.$$

The objective becomes:

$$\max_F \operatorname{Tr}(F^T X X^T F),$$

with discrete constraints on $F$.

## 9 Connection to Spectral Relaxation

If we relax the discrete constraint and allow $F$ to be any orthonormal matrix:

$$F^T F = I,$$

the solution is given by the top $k$ eigenvectors of $XX^T$.

Thus:

- K-means = constrained low-rank approximation.
- Relaxed k-means = PCA.

## 10 Conclusion

K-means can be written as:

$$\min_{H \in \{0,1\}} \min_M \|X - HM\|_F^2.$$

This shows that k-means is:

- A constrained matrix factorization problem.
- A low-rank approximation with discrete structure.

- A projection problem.
- Closely connected to PCA and spectral clustering.